Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.
The UDP protocol reveals that: The user's browser sent a UDP packet to port 53 (the DNS service port) on the DNS server at 203.0.113.2. This packet was a DNS query requesting the IP address for www.yummyrecipesforme.com. **This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message:** "UDP port 53 unreachable" (Destination port unreachable). **The port noted in the error message is used for:** Port 53, which is the standard port for DNS (Domain Name System) services. **The most likely issue is:** The DNS server is reachable at the network level, but the DNS service on that server is either not running, has crashed, or is blocked by a firewall. Because the service is not listening on port 53, the server rejects the UDP packet and returns an ICMP error instead of resolving the domain name.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.
Time incident occurred: 13:24:32 (1:24 PM), based on the timestamps in the tcpdump log. **Explain how the IT team became aware of the incident:** Multiple clients reported that they could not access www.yummyrecipesforme.com and received a "destination port

unreachable" error when trying to load the website.

Explain the actions taken by the IT department to investigate the incident:
The IT team attempted to access the website and received the same error. They then installed tcpdump to capture network traffic and reproduce the issue by trying to load the page again.

Note key findings of the IT department's investigation:

- The tcpdump log showed UDP packets being sent from the client (192.51.100.15) to the DNS server (203.0.113.2) on port 53.
- The server did not respond with a DNS answer. Instead, it returned ICMP error messages (Destination port unreachable).
- The error was consistent across multiple attempts.

Note a likely cause of the incident:
The most likely cause is that the DNS service (e.g., BIND, Unbound) on the server 203.0.113.2 has crashed or stopped running. Alternatively, a firewall rule may be blocking incoming UDP traffic on port 53, preventing the DNS service from receiving queries.